Overview

Fit-4-Life (FFL) is a fitness club company with physical locations nationwide. FFL requires a centralized database management system to manage nearly all aspects of business: its physical facilities, membership accounts, turnstile access, staff assignments, group classes, private training, locker leases, on-site retail purchases, and member billing ledgers.

This document is a narrative description of key aspects the FFL business operations.

1 Requirements

1.1 Clubs and Physical Facilities

FFL runs multiple club locations across the country. Each club is identified by a unique Club ID and records a location name, physical street address, main customer phone number, and original opening date.

Every club must employ one General Manager chosen from the staff (employee) roster. An individual staff member can serve as General Manager for at most one club at any given time.

Inside each club, members access specialized Facility Areas, such as lap pools, group fitness studios, the main gym floor, or saunas. A facility area belongs to exactly one club and is identified within that club by an area code and an area name. Each facility area also has a maximum safe occupancy limit that must be recorded.

1.2 Memberships, Turnstiles, and Emergency Contacts

When a new customer joins FFL, the new member is assigned a unique Member ID, which is used to establish the account, along with the member's full name, email address, phone number, date of birth, and enrollment date. The Member ID is based on a unique number maintained within the database. Every member must also provide exactly one primary emergency contact, which records the contact's full name, relationship to the member, and emergency phone number. An emergency contact belongs exclusively to that specific member.

FFL offers several predefined Membership Tiers (such as Basic Single Club, Signature Multi-Club, and Executive All-Access). Each tier specifies a unique tier code, tier display name, and a monthly guest pass allowance.

Monthly membership dues are location-dependent. A tier's monthly fee is determined jointly by the membership tier and the specific club location. Every valid club and tier pairing specifies a defined monthly base fee. When enrolling, every member is assigned to exactly one home club and subscribes to exactly one active membership tier.

When visiting any club, members scan their app's optical 3D barcode at the front entrance turnstile to gain entry. The system must record every check-in event, capturing the exact timestamp, the

member scanning in, and the specific club visited. A member may accumulate zero or many check-ins over time. The FFL app allows members to view all their check-ins.

1.3 Staff Personnel, Mentorship, and Personal Training

FFL employs staff across various operational roles, including front desk associates, maintenance personnel, personal trainers, and general managers. Each employee is tracked by an Employee ID, full name, hire date, hourly wage or salary, and a primary role classification. Every staff member is assigned to one home club as their primary workplace, though they may occasionally be scheduled to work shifts at other clubs.

Personal Trainers are specialized staff members who have a listed training specialty and a written bio in the system. To support staff development, senior trainers are assigned to mentor newer trainers. A senior trainer may mentor zero, one, or several junior trainers, but a junior trainer has at most one assigned mentor.

Members can purchase one-on-one private training sessions with certified trainers at an agreed session fee. Rather than paying at the time of each workout, completed training session charges are posted to the member's account ledger and billed on the 1st of the following month. A scheduled training session records the session date, time slot, duration, status (Scheduled, Completed, or No-Show), and trainer notes. Each training session pairs exactly one member with exactly one trainer. A member may book as many personal training sessions as they desire, or none at all. Although trainers like to stay busy, some often have no personal training sessions scheduled, especially for new trainers – they can go months before their first session.

1.4 Group Fitness Classes

FFL maintains a master catalog of group fitness classes (such as "Vinyasa Flow 60", "Total Body Conditioning", "Aqua Fit", etc.). Each catalog template has a unique catalog code, title, description, and an intensity rating. Based on member attendance and trainer availability, new catalog entries (classes) are added and removed from the catalog as needed.

Clubs schedule specific group class sessions based on this catalog. A scheduled session records the session date, start time, end time, the facility area where it will meet, and the trainer leading it. Each scheduled session maps back to exactly one catalog class, meets in exactly one facility area within that club, and is instructed by exactly one trainer.

Members register for scheduled class sessions in advance. A member can register for multiple classes, and a class session can accept registrations up to the capacity limit of the assigned room. The system tracks the timestamp when each registration is made and records whether the member attended or canceled. Reports are generated periodically to gauge the success of the class – attendance is monitored to ensure only the most well-attended classes are scheduled. If a scheduled class session consistently has zero or very low attendance by members, it will not be scheduled in the future.

1.5 Dedicated Locker Rentals

Clubs provide premium lockers that members can optionally rent on a monthly basis; members can cancel the rental anytime and the fee will be prorated for the rest of the month. Each locker has a locker number, a mechanical or digital combination code, and a locker room location description (e.g., "Men's Locker Room A"). A locker belongs to a specific club. If a premium locker is rented by a member, the member has exclusive access to the locker.

The base rate of a premium locker is on a club by club basis. Each club has, potentially, a different base rate. A member may rent no more than one locker across the entire company (all club locations). Fees are also prorated for the first month, then added as a monthly locker rental fee. All fees (prorated or not) are included in the member's account ledger.

1.6 On-Site Pro Shops and House Tabs

Many, but not all, clubs host an on-site retail "pro shop" (store) selling healthy prepared foods, smoothies, supplements, apparel, and workout gear.

If a club has a pro shop, details regarding the pro shop are maintained, such as: store name, square footage, and operating hours).

The FFL company maintains a retail merchandise inventory where each product has an item code, item name, category, and standard retail price. Individual clubs stock varying quantities of these merchandise items.

Members can purchase items at a club store either by paying on the spot or by scanning their member barcode to charge the purchase directly to their Member Tab. When a tab purchase occurs, the system records the transaction timestamp, the club store where the purchase took place, the specific items and quantities purchased, and the member charging the tab. Items charged to a member's tab are added to their outstanding ledger balance.

1.7 Billing, Invoicing, and Payments

On the 1st of every calendar month, the billing engine assesses all one-time and recurring charges for every active member, generating a monthly fee assessment (invoice) that details the billing date, the billing period covered, and the total balance due. The assessment itemizes and aggregates: (1) recurring base tier dues, (2) monthly locker rental fees, (3) personal training sessions conducted during that period, and (4) any unpaid house tab charges accrued at club pro shops.

Members can pay balances by cash, credit card, or ACH bank transfer. Cash payments must be performed in person at a club front desk, and the system must record the FFL employee who processed them. Credit card and ACH payments may be made online or in person.

Members can securely save payment methods to their profile for future use:

- **Saved Credit Cards:** Store cardholder name, processor card token, expiration month/year, and billing postal code.

- **Saved ACH Accounts:** Store bank name, account holder name, account type (Checking or Savings), bank routing number, and masked bank account number.

A member may designate at most one of their saved payment methods as their active Auto-Pay method for automatic monthly dues. A saved method does not have to be enrolled in auto-pay.

Every financial transaction processed is logged as a payment transaction, capturing the payment timestamp, amount paid, transaction status (Success, Failed, or Pending), the payment method type used ("Cash", "Credit Card", or "ACH"), and which staff member processed the payment (if paid in cash/in person). A payment transaction can be funded using a saved payment method on file, an ad-hoc ("transient") card or bank account entered for a single transaction without saving, or in-person cash.

In the club ledger, a single payment transaction can settle multiple outstanding fee assessments, and an individual fee assessment can be paid down over time through multiple installments. Members can review their complete payment and invoice history at any time. For charges listed on the invoice history, all pertinent details are line-itemed and include enough detail to allow the member to identify the charges: for example, if the charge is from a pro shop, the member should be able to view relevant details.